

1) Design power = Service factor $\times$ drive power

From table A-2:

The service factor is 1.1.

$$\therefore \text{Design power} = 1.1 \times 7.5 \\ = 8.25 \text{ W}$$

From table A-3, using the drive speed of 1160 rpm, the V-belt cross section is SPZ.

The tentative centre distance (TCD) is given as 760 mm.

Finding the tentative belt length (TBL)

$$\begin{aligned} \text{TBL} &= 2 \times \text{TCD} + 1.57(D_2 + D_1) + \frac{(D_2 - D_1)^2}{4 \times \text{TCD}} \\ &= 2(760) + 1.57(200 + 125) + \frac{(200 - 125)^2}{4 \times 760} \\ &= 2032.100329 \text{ mm} \end{aligned}$$

From Table A-1, selecting the closest belt length:

$$\text{Belt length} = 2000 \text{ mm}$$

1) Calculating the actual centre distance:

$$\begin{aligned} B &= 4L - 6.28(D_2 + D_1) \\ &= 4(2000) - 6.28(200 + 125) \\ &= 5959 \text{ mm} \end{aligned}$$

$$\begin{aligned} C &= \frac{B + \sqrt{B^2 - 32(D_2 - D_1)^2}}{16} \\ &= \frac{5959 + \sqrt{5959^2 - 32(200 - 125)^2}}{16} \\ &= 743.9298504 \\ &\approx 744 \text{ mm} \end{aligned}$$

Determining the angle of contact:

$$\begin{aligned} \theta_1 &= 180^\circ - 2 \arcsin \left(\frac{D_2 - D_1}{2C} \right) \\ &= 180^\circ - 2 \arcsin \left(\frac{200 - 125}{2(744)} \right) \\ &= 179.8991413^\circ \\ &\approx 180^\circ \end{aligned}$$

From table A-5:

$$C_\theta = 1.00$$

1) From table A-6:

$$C_L = 1.04$$

Speed ratio:

$$SR = \frac{200}{125} \\ = 1.6$$

From Table A-7a, interpolating:

$$\frac{1.6 - 1.5}{3.00 - 1.5} = \frac{RP - 3.22}{3.25 - 3.22}$$

$$RP = 3.222 \text{ kW belt}^{-1}$$

Determining the corrected rated power (CRP):

$$\begin{aligned} CRP &= C_0 C_L RP \\ &= 1.00(1.04)(3.222) \\ &= 3.35088 \text{ W} \end{aligned}$$

$$\text{Number of belts} = \frac{8.25}{3.35088}$$

$$= 2.462639822$$

$$\approx 3$$

$\therefore$ The belt drive requires 3 belts of SPZ cross-section and length of 2000mm (shown).

2) Design power = service factor $\times$ drive power

From table A-2:

$$\text{Service factor} = 1.0$$

$$\begin{aligned}\text{Design power} &= 1.0 \times 40 \\ &= 40 \text{ kW}\end{aligned}$$

From table A-3:

V-belt cross-section is SPA.

Given a belt speed, $v_b = 14 \text{ m s}^{-1}$

$$v_b = \frac{D_1 \omega_1}{2}$$

$$\begin{aligned}D_1 &= \frac{2v_b}{\frac{2\pi n_1}{60}} = \frac{60v_b}{\pi n_1} \\ &= \frac{60(14)}{\pi(1200)}\end{aligned}$$

$$= 0.2228169203 \text{ m}$$

From Table A-4:

Select $D_1 = 200 \text{ mm}$ to not exceed v_b

Checking the belt speed:

$$v_b = \frac{0.200(2\pi)(1200)}{2(60)}$$

$$= 4\pi \approx 12.57 \text{ m s}^{-1}$$

2) Speed ratio = 1.5

$$1.5 = \frac{D_2}{D_1}$$

$$D_2 = 1.5(200) \\ = 300 \text{ mm}$$

From Table A-4:

$$D_2 = 315 \text{ mm}$$

Check if n_2 is within $780 \pm 5\%$ rpm, i.e.

$$741 \text{ rpm} \leq n_2 \leq 819 \text{ rpm}$$

$$n_2 = \frac{D_1 n_1}{D_2} \\ = \frac{200(1200)}{315}$$

$$= 761.9047619 \text{ rpm (within range)}$$

$$2) \text{ Actual speed ratio} = \frac{315}{200} = 1.575$$

From Table A-7b, interpolating:

$$RP = \frac{1.575 - 1.5}{3.00 - 1.5} (8.59 - 8.51) + 8.51$$

$$= \frac{4257}{500} \text{ kW belt}^{-1}$$

Corrected rated power (CRP):

$$CRP = C_0 C_L RP$$

$$= 0.48 \times 0.94 \left(\frac{4257}{500} \right)$$

$$= 7.8430968 \text{ kW belt}^{-1}$$

$$\text{Number of belts} = \frac{40}{7.8430968}$$

$$= 5.100026306 \approx 6$$

$\therefore$ The V-belt cross section is SPA and 6 belts are required.